Functions of Several Variables & Partial Derivatives

ANSWER KEY

Section 1: Domain, Range, and Level Curves

1 Find the domain of the function $f(x, y) = \ln(9 - x^2 - y^2)$.

a. **The interior of the circle** $x^2 + y^2 < 9$ ✓

b. The exterior of the circle $x^2 + y^2 > 9$

c. The closed disk $x^2 + y^2 \leq 9$

d. All of $\mathbb{R}^2$

The argument of the natural logarithm must be strictly positive: $9 - x^2 - y^2 > 0 \Leftrightarrow x^2 + y^2 < 9$, which is the open interior of a circle of radius 3 centered at the origin.

2 What geometric curves are the level curves of $z = x^2 + y^2$ for $c > 0$?

a. **Concentric circles centered at** $(0, 0)$ ✓

b. Lines of slope 1

c. Parabolas opening upward

d. Hyperbolas

Setting $z = c \Rightarrow x^2 + y^2 = c$. For $c > 0$, this is a family of concentric circles of radius $\sqrt{c}$.

3 What is the range of $g(x, y) = \sqrt{16 - x^2 - y^2}$?

a. $[0, 4]$ ✓

b. $[0, 16]$

c. $[-4, 4]$

d. $[0, \infty)$

When $x^2 + y^2 = 0$, $g(0, 0) = \sqrt{16} = 4$. On the boundary $x^2 + y^2 = 16$, $g = 0$. Since square roots are non-negative, the range is $[0, 4]$.

Section 2: First-Order Partial Derivatives

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4 Find the partial derivative $f_x(x, y)$ for $f(x, y) = 3x^3y^2 - 5x^2 + 4y^3$.

a. $9x^2y^2 - 10x$ ✓

b. $6x^3y + 12y^2$

c. $9x^2y^2 - 10x + 12y^2$

d. $3x^2y^2 - 10x$

Treat y as a constant: $\frac{\partial}{\partial x}[3x^3y^2] = 3(3x^2)y^2 = 9x^2y^2$; $\frac{\partial}{\partial x}[-5x^2] = -10x$; $\frac{\partial}{\partial x}[4y^3] = 0$. Result: $9x^2y^2 - 10x$.

5 Find $f_y(x, y)$ for $f(x, y) = e^{2x}\sin(3y)$.

a. $3e^{2x}\cos(3y)$ ✓

b. $2e^{2x}\sin(3y)$

c. $6e^{2x}\cos(3y)$

d. $-3e^{2x}\cos(3y)$

Treat x as constant: $\frac{\partial}{\partial y}[e^{2x}\sin(3y)] = e^{2x} \cdot 3\cos(3y) = 3e^{2x}\cos(3y)$.

6 Evaluate $f_x(1, 2)$ for $f(x, y) = x^2y + 3xy^2$.

a. 16 ✓

b. 28

c. 12

d. 8

$f_x(x, y) = 2xy + 3y^2$. At $(1, 2)$: $f_x(1, 2) = 2(1)(2) + 3(2^2) = 4 + 3(4) = 4 + 12 = 16$.

Section 3: Higher-Order Partials and Clairaut's Theorem

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7 What does Clairaut's Theorem state about mixed second-order partial derivatives f_{xy} and f_{yx}?

a. **If both f_{xy} and f_{yx} are continuous on an open region, then $f_{xy} = f_{yx}$** ✓

b. Mixed partials are always equal to 0

c. The sum $f_{xy} + f_{yx} = 0$

d. Equality holds only for polynomials

Clairaut's Theorem guarantees that mixed partial derivatives are equal ($f_{xy} = f_{yx}$) provided they are continuous on an open disk.

8 For $f(x, y) = x^3y^4$, verify Clairaut's Theorem by finding f_{xy}.

a. $12x^2y^3$ ✓

b. $3x^2y^4$

c. $4x^3y^3$

d. $6xy^2$

$f_x = 3x^2y^4$. Then $f_{xy} = \frac{\partial}{\partial y}[3x^2y^4] = 3x^2(4y^3) = 12x^2y^3$.

9 Find f_{xx} for $f(x, y) = x^4 - 2x^2y^3 + y^5$.

a. $12x^2 - 4y^3$ ✓

b. $4x^3 - 4xy^3$

c. $12x^2$

d. $-12x^2y^2$

$f_x = 4x^3 - 4xy^3$. Differentiating again with respect to x: $f_{xx} = 12x^2 - 4y^3$.

Section 4: Tangent Planes and Linear Approximations

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10 What is the equation of the tangent plane to the surface $z = f(x, y)$ at the point (x_0, y_0, z_0)?

a. $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$ ✓

b. $z - z_0 = f_x(x - x_0) - f_y(y - y_0)$

c. $z = f_x x_0 + f_y y_0$

d. $(x - x_0) + (y - y_0) + (z - z_0) = 0$

The linear approximation (tangent plane) formula is $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$.

11 Find the equation of the tangent plane to $z = x^2 + y^2$ at $(1, 2, 5)$.

a. $z - 5 = 2(x - 1) + 4(y - 2) \Leftrightarrow 2x + 4y - z = 5$ ✓

b. $2x + 4y + z = 15$

c. $x + 2y - z = 0$

d. $4x + 2y - z = 5$

$f_x = 2x \Rightarrow f_x(1, 2) = 2$. $f_y = 2y \Rightarrow f_y(1, 2) = 4$. Plane:
$z - 5 = 2(x - 1) + 4(y - 2) \Rightarrow z - 5 = 2x - 2 + 4y - 8 \Rightarrow 2x + 4y - z = 5$.

12 What is a normal vector to the surface $z = f(x, y)$ at (x_0, y_0)?

a. $\mathbf{n} = [f_x(x_0, y_0), f_y(x_0, y_0), -1]$ ✓

b. $\mathbf{n} = [f_x, f_y, 1]$

c. $\mathbf{n} = [1, 1, 1]$

d. $\mathbf{n} = [-f_y, f_x, 0]$

Rewriting as $F(x, y, z) = f(x, y) - z = 0$, the gradient $\nabla F = [f_x, f_y, -1]$ forms an outward normal vector.

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